

A Large-Tandem Model for an Unresolved Canon-Avoidance Sequence

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Abstract

Let b_n count canon words in which each letter $1, \dots, n$ occurs five times, the common underlying permutation is $12 \cdots n$, and the patterns 122221 and 211112 are avoided. We give a bijection from these words to quarter-plane excursions of length $5n$ with step set

$$\{(1, 0), (-2, 2), (0, -1)\}.$$

This identifies the sequence with a large-tandem model. Applying Wallner's general theorems shows that $B(z) = \sum_{n \geq 0} b_n z^n$ is not D-finite, that (b_n) is not P-recursive, and that

$$b_n = \Theta\left(\left(\frac{3125}{16}\right)^n n^{-1-\pi/\arccos(2/3)}\right).$$

Exact integer computations give four further terms, through b_{11} . These results are subsidiary: they do not provide a solved univariate generating function, closed form, or unrestricted exact recurrence of the kind requested in the original enumeration problem, which therefore remains unresolved.

1 The canon-word problem

A *canon word* is a regular word for which the permutation obtained by reading the first occurrences of the letters is the same as that obtained by reading the second occurrences, and so on. Laudone introduced the relevant pattern-avoidance framework and isolated the case considered here as an open enumeration problem [1]. We use his reduction with fixed underlying permutation. Pattern containment is classical subsequence containment for words, preserving both equalities and strict inequalities between letters. Since the two forbidden patterns comprise their full two-letter relabeling orbit, the corresponding count over all underlying permutations is $n!b_n$.

For $n \geq 1$, let P_n have elements (i, r) with $1 \leq i \leq n$ and $1 \leq r \leq 5$. Its order is generated by

$$(i, r) < (i, r+1) \quad (1 \leq r < 5), \quad (1)$$

$$(i, r) < (i+1, r) \quad (1 \leq i < n), \quad (2)$$

$$(i-1, 5) < (i, 4), \quad (i-1, 2) < (i, 1) \quad (2 \leq i \leq n). \quad (3)$$

The omitted ranges in (1) and (2) are the natural ones. Relations (1)–(2) encode the occurrence order and the identity underlying permutation. Laudone's avoidance criterion [1, Lemma 5.4],

specialized to the parameters $j = 4$ and $k = 5$, gives exactly (3). Reading a linear extension chronologically therefore recovers a unique avoiding canon word, and

$$b_n = e(P_n). \quad (4)$$

We put $b_0 = 1$ when using generating functions. This convention does not alter the source problem, which is stated for $n \geq 1$.

The contribution below is the explicit identification of (4) with a large-tandem excursion model. Wallner developed that model and proved general asymptotic and non-D-finiteness theorems for it [2]. A separate recent permutation-enumeration study uses the same broad combination of a generating function, analytic coefficient estimates, and exact computation, but for a different sequence [3]; none of its results is used here.

2 Reduction to three chains and quadrant excursions

Define three ordered lists of elements of P_n by

$$A_{2i-1} = (i, 1), \quad A_{2i} = (i, 2), \quad B_i = (i, 3), \quad C_{2i-1} = (i, 4), \quad C_{2i} = (i, 5).$$

Lemma 1 (Three-chain presentation). *The order on P_n is generated equivalently by the chain orders*

$$A_1 < \cdots < A_{2n}, \quad B_1 < \cdots < B_n, \quad C_1 < \cdots < C_{2n},$$

together with the cross-links

$$A_{2i} < B_i < C_{2i-1} \quad (1 \leq i \leq n). \quad (5)$$

Proof. The within-letter relations between rows 1 and 2, together with $(i-1, 2) < (i, 1)$, interleave those rows as

$$(1, 1) < (1, 2) < (2, 1) < (2, 2) < \cdots < (n, 1) < (n, 2).$$

Thus the A elements form the displayed chain. The same argument with rows 4 and 5 and the other relation in (3) gives the C chain; the row-3 relations give the B chain. The remaining adjacent-row generators are $(i, 2) < (i, 3) < (i, 4)$, which are precisely (5). Conversely, every generator (1)–(3) lies in the transitive closure of these three chains and cross-links. Hence the two presentations define the same poset. $\square$

Theorem 2 (Excursion bijection). *For every $n \geq 0$, b_n is the number of excursions of length $5n$ from the origin to the origin in $\mathbb{N}^2$ with steps*

$$(1, 0), \quad (-2, 2), \quad (0, -1). \quad (6)$$

Proof. Within each chain of Lemma 1, the internal order is forced. Recording only the chain name along a linear extension is consequently injective and reversible. If a prefix contains a, b, c elements from the A, B, C chains, respectively, then the cross-links say exactly that

$$a \geq 2b \geq c. \quad (7)$$

Indeed, B_i is available only after A_{2i} , while C_{2i-1} is available only after B_i ; the chain orders handle the even-indexed C elements. Thus linear extensions are in bijection with words having $2n, n, 2n$ letters of the three types and satisfying (7) in every prefix.

Apply the gap map

$$(a, b, c) \longmapsto (u, v) = (a - 2b, 2b - c).$$

Condition (7) becomes $(u, v) \in \mathbb{N}^2$, and appending an A , B , or C gives, in that order, the three steps in (6). The endpoint $(2n, n, 2n)$ maps to the origin.

Conversely, suppose an excursion uses the three steps with multiplicities r, s, t . Returning to the origin forces $r = 2s$ and $t = 2s$. If its length is $5n$, then $(r, s, t) = (2n, n, 2n)$, and the inverse gap construction gives a unique word satisfying (7), hence a unique linear extension. The empty excursion accounts for $n = 0$. $\square$

This also gives a uniform exact dynamic program. Let $q_m(u, v)$ count walks of length m ending at $(u, v) \in \mathbb{N}^2$, and set it to zero outside the quadrant. Then

$$\begin{aligned} q_0(u, v) &= \mathbf{1}_{(u,v)=(0,0)}, \\ q_{m+1}(u, v) &= q_m(u-1, v) + q_m(u+2, v-2) + q_m(u, v+1), \\ b_n &= q_{5n}(0, 0). \end{aligned} \tag{8}$$

For reference, let

$$Q(x, y; t) = \sum_{m,u,v \geq 0} q_m(u, v) x^u y^v t^m.$$

A step decomposition yields the exact catalytic functional equation

$$\begin{aligned} Q(x, y; t) &= 1 + txQ(x, y; t) \\ &\quad + t \frac{y^2}{x^2} (Q(x, y; t) - Q(0, y; t) - x[x^1]Q(x, y; t)) \\ &\quad + t \frac{1}{y} (Q(x, y; t) - Q(x, 0; t)). \end{aligned} \tag{9}$$

If $E(t) = Q(0, 0; t)$ and $B(z) = \sum_{n \geq 0} b_n z^n$, periodicity gives

$$E(t) = B(t^5). \tag{10}$$

Equations (8)–(10) are exact computational characterizations. They are not presented as a solved univariate generating function or as a closed form for b_n .

3 Non-D-finiteness and asymptotics

The step set (6) is Wallner's large-tandem model with parameters $(A, B, C) = (1, 2, 1)$ [2]. We record the consequences in the precise form needed here.

Theorem 3. *The ordinary generating function $B(z) = \sum_{n \geq 0} b_n z^n$ is not D-finite. Equivalently, (b_n) is not P-recursive. Moreover,*

$$b_n = \Theta\left(\left(\frac{3125}{16}\right)^n n^{-1-\pi/\arccos(2/3)}\right). \tag{11}$$

No value of the leading constant is asserted.

Proof. For a large-tandem model, Wallner's parameters are

$$\gamma = -\frac{B}{\sqrt{(A+B)(B+C)}}, \quad \alpha = -1 - \frac{\pi}{\arccos(-\gamma)}.$$

Here $\gamma = -2/3$, so $\alpha = -1 - \pi/\arccos(2/3)$. The non-D-finiteness criterion in [2, Theorem 3.3] applies because

$$\frac{B^2}{(A+B)(B+C)} = \frac{4}{9} \notin \left\{ \frac{1}{4}, \frac{1}{2}, \frac{3}{4} \right\}.$$

Its other structural hypotheses can be checked directly: the parameters are coprime; the steps span the plane and have the strictly positive relation

$$2(1, 0) + (-2, 2) + 2(0, -1) = (0, 0);$$

$(1, 0)$ is nonnegative; and $(2, 1)$ reaches the origin by one $(-2, 2)$ step followed by three $(0, -1)$ steps while staying in the quadrant. A return has multiplicities $(2n, n, 2n)$, and the word $AABCC$ is a return of length five, so the period is exactly five.

At the positive critical point

$$X = 2^{1/5}, \quad Y = 2^{-1/5},$$

the step polynomial $S(x, y) = x + (y/x)^2 + y^{-1}$ has value

$$\mu = S(X, Y) = \frac{5}{2^{4/5}}, \quad \mu^5 = \frac{3125}{16}.$$

Wallner's periodic excursion estimate [2, Theorem 2.3 and Lemma 3.2] now gives (11); the constant factor 5^α is absorbed by Θ .

The same work proves that the full excursion series $E(t)$ is not D-finite. If B were D-finite, closure under polynomial substitution would make $B(t^5) = E(t)$ D-finite, contradicting (10). Finally, the coefficient characterization of D-finite series identifies D-finiteness of B with P-recursiveness of (b_n) . $\square$

Remark 4. The theorem rules out linear recurrences with polynomial coefficients. It does not rule out every possible exact recurrence, and in particular it does not solve the original request for an exact univariate enumeration valid for all n .

4 Exact coefficients and reproducibility

Recurrence (8), evaluated with integer arithmetic, gives

n	b_n
0	1
1	1
2	19
3	795
4	48875
5	3780362
6	340746821
7	34264303289
8	3740554409595
9	435337019298913
10	53331481655131738
11	6813743109121574171

The values through b_7 agree with Laudone’s list [1, Question 5.12]. The four subsequent values were checked by two exact encodings: the quadrant recurrence (8) and a generator-level order-ideal dynamic program built directly from (1)–(3). The value of b_8 was also checked by a third program using five row-prefix coordinates. All programs use arbitrary-precision integers; no floating-point inference enters any theorem. The recurrence (8) is sufficient to reproduce the table without relying on those implementations.

The closest inspected source is Laudone’s formulation and open question; the general large-tandem consequences come from Wallner. A bounded literature search found no inspected source explicitly making the connection of Theorem 2. This is a limited comparison, not a claim of priority or an exhaustive statement about the literature.

Computational and AI assistance. OpenAI Codex was used to assist with structural exploration, preparation of exact-integer verification programs, consistency checks, and drafting. The three-chain proof and the applications of the cited theorems were then stated explicitly in the manuscript and checked against the frozen definitions and the cited primary sources. The software checks certify only the finite computations they encode; they do not constitute a proof of the all- n claims or independent peer review.

References

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